

A negative answer to the cusp finite-subset-retraction question in arXiv:2004.09419 via arXiv:2108.03535

Literature-status note

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Status: literature-implied negative answer. The cusp has no Lipschitz retraction $X(4) \rightarrow X(3)$, hence it cannot have the family $X(n) \rightarrow X(n-1)$ requested in the source. This is a direct identification with a later theorem, not a new result.

1 The source question

Let

$$X = \{(s, |s|^{1/2}) : s \in [-1, 1]\} \subset \mathbb{R}^2,$$

with the restricted Euclidean metric. Question 5.4.1 of arXiv:2004.09419 (PDF page 131, printed page 122), immediately after Example 5.4.6, asks:

If X is the cusp in Example 5.4.6, do we have Lipschitz retractions $X(n) \rightarrow X(n-1)$?

Here $X(n)$ is the space of nonempty subsets of X of cardinality at most n , with the Hausdorff metric [1].

2 The later obstruction

Kovalev's Theorem 5.1 in arXiv:2108.03535v2 (statement on PDF page 10) says that if a metric space Y supports an L -Lipschitz retraction $Y(4) \rightarrow Y(3)$ and $[0, 1]$ admits an L -bi-Lipschitz embedding into Y , then there are $r, M > 0$ such that every $p, q \in Y$ with $d(p, q) \leq r$ can be joined in Y by a curve of length at most $Md(p, q)$ [2]. Thus one retraction $Y(4) \rightarrow Y(3)$ already forces uniform local quasiconvexity whenever Y contains a bi-Lipschitz interval.

Both facts needed to apply the contrapositive are elementary for the cusp. First, the positive branch contains a bi-Lipschitz interval: for $\gamma(u) = (u^2, u)$, $0 \leq u \leq 1$,

$$|u - v| \leq \|\gamma(u) - \gamma(v)\| = |u - v| \sqrt{1 + (u + v)^2} \leq \sqrt{5} |u - v|.$$

Second, X is not uniformly locally quasiconvex at its vertex. Put

$$p_t = (-t^2, t), \quad q_t = (t^2, t), \quad 0 < t \leq 1.$$

Their ambient distance is $2t^2$. Every curve in X joining the two branches must pass through $(0, 0)$, so its length is at least

$$\|p_t\| + \|q_t\| = 2t\sqrt{1 + t^2}.$$

The ratio of this lower bound to $\|p_t - q_t\|$ is $\sqrt{1+t^2}/t \rightarrow \infty$. Therefore no constants r, M as in Theorem 5.1 can exist.

It follows contrapositively that there is no Lipschitz retraction $X(4) \rightarrow X(3)$. In particular, the answer to Question 5.4.1 is

No.

3 Provenance and scope

Kovalev defines the Lipschitz Clustering Property by the existence of all consecutive retractions $Y(n) \rightarrow Y(n-1)$; this is the same property asked about here. His paper does not cite or name Akofof's cusp question in the statement or discussion of Theorem 5.1. The decisive relation is therefore an agent-identified implication of a later published theorem, appropriately recorded as a literature-implied answer. No assertion is made here about Akofof's other FSR questions.

References

- [1] E. Akofof, *Metric Geometry of Finite Subset Spaces*, arXiv:2004.09419 (2020), Question 5.4.1.
- [2] L. V. Kovalev, *Lipschitz clustering in metric spaces*, arXiv:2108.03535v2 (2021), Theorem 5.1; J. Geom. Anal. 32 (2022).